

Literature resolution: cyclic polynomials on the bidisk

ARXIV:1310.4094 | ANSWERED BY ARXIV:1408.3616

STATUS Complete later-literature answer	SCOPE Source Problem 5.2
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Source question

Problem 5.2 of arXiv:1310.4094 asks for a characterization, for every α at most 1, of cyclic polynomials in the product-weight Dirichlet-type space on the bidisk.

Exact later answer

ArXiv:1408.3616 explicitly names the source's Problem 5.2 and says: *In the present paper, we solve this problem and provide a complete characterization.* Its Main Theorem applies to an irreducible polynomial f with no zero in the open bidisk.

Parameter	Classification
$\alpha \leq 1/2$	Every such irreducible polynomial is cyclic.
$1/2 < \alpha \leq 1$	Cyclic exactly when the torus zero set is empty or finite, or the polynomial is a constant multiple of z_1 or z_2 for a unimodular z .
$\alpha > 1$	Cyclic exactly when the torus zero set is empty.

How to use the theorem

A general polynomial is cyclic exactly when each irreducible factor is cyclic, because polynomials are multipliers. Any zero in the open bidisk rules out cyclicity before the boundary classification is applied.

Why the identification is exact

Both papers use the same coefficient norm: the squared norm is the sum over $k, l \geq 0$ of $(k+1)$ raised to α times $(l+1)$ raised to α , times the squared modulus of the coefficient $a(k, l)$. The 2014 theorem is therefore an answer in the identical space, not an analogy using a different Dirichlet scale.

Boundary of this packet

This resolves only the polynomial-classification problem. The preceding Brown-Shields question concerns arbitrary outer functions and is not claimed to be settled here.

References

[1] C. Beneteau, A. A. Condori, C. Liaw, D. Seco, and A. A. Sola, *Cyclicity in Dirichlet-type spaces and extremal polynomials II: functions on the bidisk*, arXiv:1310.4094.

[2] C. Beneteau, G. Knese, L. Kosinski, C. Liaw, D. Seco, and A. A. Sola, *Cyclic polynomials in two variables*, arXiv:1408.3616, Main Theorem.